

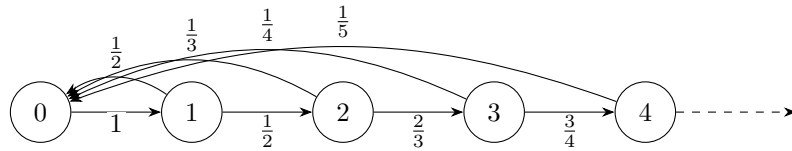
Discussion section 6

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1 Chp 3, Q27

Consider the infinite Markov chain on the non-negative integers described by the following figure.



- Show that the chain is irreducible and aperiodic.
- Show that the chain is recurrent.
- Show that the chain is null recurrent by computing the first return time to 0 for the chain started at 0.

Proof. This is a homework problem.

- If $i < j$, one can reach from state i to state j by $i \rightarrow i+1 \rightarrow \dots \rightarrow j-1 \rightarrow j$, and reach from state j to state i by $j \rightarrow 0 \rightarrow 1 \rightarrow 2 \rightarrow \dots \rightarrow i-1 \rightarrow i$. Thus the chain is irreducible.

Since periodicity is a class property, it suffices to look at the state 0. By looking at the two paths $0 \rightarrow 1 \rightarrow 0$ and $0 \rightarrow 1 \rightarrow 2 \rightarrow 0$, and $\gcd(2, 3) = 1$, we have state 0 is aperiodic. Therefore the whole chain is aperiodic.

- Since recurrence is a class property, it suffices to show state 0 is recurrent. By looking at the graph, if we want to return from 0 to 0 in k -steps, the only possible path is $0 \rightarrow 1 \rightarrow 2 \rightarrow \dots \rightarrow k-2 \rightarrow k-1 \rightarrow 0$, with the probability $1 \cdot \frac{1}{2} \cdot \frac{2}{3} \cdot \dots \cdot \frac{k-2}{k-1} \cdot \frac{1}{k} = \frac{1}{(k-1)k}$.

In fact the hint is misleading. It should be moved to (c) and I've done it. Let's just start from definition, computing the following quantity:

$$\mathbb{P}(T_0 < \infty | X_0 = 0) = \sum_{k=1}^{\infty} \mathbb{P}(T_0 = k | X_0 = 0) = \sum_{k=2}^{\infty} \frac{1}{(k-1)k} = 1,$$

where T_0 is the first returning time to state 0. By definition, state 0 is recurrent.

- (c) Null recurrency is a class property. It suffices to show state 0 is null recurrent. We used the previous computation, computing the expected returning time:

$$\begin{aligned} \mathbb{E}[T_0 | X_0 = 0] &= \sum_{k=1}^{\infty} k \cdot P(T_0 = k | X_0 = 0) \\ &= \sum_{k=2}^{\infty} k \cdot \frac{1}{(k-1)k} = \sum_{k=2}^{\infty} \frac{1}{k-1} = \infty. \end{aligned}$$

By definition, state 0 is null recurrent.

□

2 Chp 3, Q37

Show that all two-state Markov chains, except for the trivial chain whose transition matrix is the identity matrix, are time reversible.

Proof. Recall the transition matrix of a two-state Markov chain is

$$P = \begin{matrix} & \begin{matrix} 1 & 2 \end{matrix} \\ \begin{matrix} 1 \\ 2 \end{matrix} & \begin{pmatrix} 1-p & p \\ q & 1-q \end{pmatrix} \end{matrix}.$$

with stationary distribution $\pi = (\frac{q}{p+q}, \frac{p}{p+q})$, if $p \neq 0$ and $q \neq 0$. We need to check the detailed balanced equations are satisfied, i.e.,

$$\pi_i P_{ij} = \pi_j P_{ji}$$

for all $i, j \in \{1, 2\}$.

□

This is equivalent to

$$\pi_1 P_{12} = \pi_2 P_{21},$$

i.e.,

$$\frac{q}{p+q} \cdot p = \frac{p}{p+q} \cdot q.$$

Thus the balanced equations are always satisfied, except for the trivial chain since the trivial chain is not irreducible.

3 Chp 3, Q43

A markov chain has transition matrix

$$P = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{pmatrix} 1/2 & 1/4 & 0 & 1/4 \\ p & 0 & 1-p & 0 \\ 0 & 1/4 & 1/2 & 1/4 \\ q & 0 & 1-q & 0 \end{pmatrix} \end{matrix}$$

- (a) For what values of p and q is the chain ergodic?
- (b) For what values of p and q is the chain reversible?

Proof. (a) Draw out the graph. The edge case will be $p, q \in \{0, 1\}$, which will affect the path connectedness of the graph, or equivalently, irreducibility of the Markov chain. By analyzing separately the case where $p, q \in \{0, 1\}$, the chain is irreducible unless $p = q = 0$ or $p = q = 1$. Since $P_{11} = 1/2 > 0$, the chain is ergodic unless $p = q = 0$ or $p = q = 1$.

- (b) By definition, we first need to make sure the chain is irreducible, i.e., $p = q = 0$ and $p = q = 1$ should be ruled out.

Then we need to check the detailed balanced equations are satisfied, i.e.,

$$\pi_i P_{ij} = \pi_j P_{ji}$$

for all $i, j \in \{1, 2, 3, 4\}$, which is the following system:

$$\begin{cases} \pi_1 \cdot \frac{1}{4} = \pi_2 \cdot p, \\ \pi_1 \cdot 0 = \pi_3 \cdot 0, \\ \pi_1 \cdot \frac{1}{4} = \pi_4 \cdot q, \\ \pi_2 \cdot (1-p) = \pi_3 \cdot \frac{1}{4}, \\ \pi_2 \cdot 0 = \pi_4 \cdot 0, \\ \pi_3 \cdot \frac{1}{4} = \pi_4 \cdot (1-q). \end{cases} \quad (3.1)$$

The first three equations give

$$\frac{1}{4}\pi_1 = \pi_2 p = \pi_4 q. \quad (3.2)$$

The last three equations give

$$\frac{1}{4}\pi_3 = \pi_2(1-p) = \pi_4(1-q). \quad (3.3)$$

Adding up (3.2) and (3.3) we have $\pi_2 = \pi_4$, and thus $p = q$.

In fact $p = q$, with $0 < p < 1$ are the only constraints. Using $p = q$, together with the normalization condition $\pi_1 + \pi_2 + \pi_3 + \pi_4 = 1$, we can solve for the stationary distribution $\pi = (\frac{2}{3}p, \frac{1}{6}, \frac{2}{3}(1-p), \frac{1}{6})$.

The answer is any $p = q$ with $0 < p < 1$.

□